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Roll No

EX-603 (GS)
B.E. VI Semester Examination, June 2020
Grading System (GS)
Switch Gear and Protection
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Describe construction, operating principle and application of vacuum circuit breaker for what voltage range is it recommended.
2. Describe the construction, principle of operation and application of an
 - a) Induction disc
 - b) Induction cup type of relayWhat is ratio of reset to pickup value in case of these relays?
3. What are the required features of an ideal protective relay? Define terms:
 - i) Operating time
 - ii) Seal in relay
 - iii) Burden
4. Discuss the protection of a three-phase alternator in the event of following:
 - i) Loss of prime mover
 - ii) Loss of excitation
5. Calculate the symmetrical components of the following unbalanced line to neutral voltage $V_a = 100 \angle 90^\circ$, $V_b = 116 \angle 0^\circ$, $V_c = 71 \angle 224.8^\circ$.

OR

Explain the working principle, construction and merits of minimum oil C.B. with the help of neat sketch.
6. What is a biased differential protection. Why it is used. Draw a neat circuit diagram of differential protection for Transformer and explain how it operation.

OR

Describe different protection of generators.
7. The line of ground voltages on the high voltage side of a step up transformer are 100kV, 33kV and 38kV on phase a, b and c respectively. The voltage of phase a leads that phase b by 100° and lags that of phase c by 176.5° . Determine analytically the symmetrical components of voltages.
8. Explain the following:(Any two)
 - i) Surge absorber and diverters.
 - ii) Valve type lightning arrestors.
 - iii) Insulation co-ordination.
 - iv) Phenomena of lightning.
